

Meeting 4

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Date:	8 May 2025
Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
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Outcome of previous meeting

- Look into the truncation-termination difference w.r.t. GAE in CleanRL and Stable Baselines 3
- Run CartPole-v1 with observation normalization and reward normalization (as well as transformation) wrappers
- Apply the same experiment of CartPole-v1 and HalfCheetah-v4 on Ant-v4, as the harder environment
- Repeat the procedure of applying the default hyperparameters of baseline frameworks on each other for SAC

Results during the last weeks

- Last results were too optimistic: I fixed a bug with wrong network initialization (due to making it accessible for easy change using a YAML file) in CleanRL. This caused the experiments to be faulty.
- Stable Baselines 3 handles truncation by $r_t \leftarrow r_t + \gamma V^\phi(s_{t+1})$, where r_t is the reward at time t , γ the discount factor and $V^\phi(s_{t+1})$ the expected value of the truncated state s_{t+1} according to value function V^ϕ , parameterized by ϕ . CleanRL has no bootstrapping.
- Reproduction of the original PPO experiments shows differences, a few of which can be traced back to the truncation-termination problem. Others may be changes in physics engine MuJoCo (within integration), and possible misinterpretation of the use of learning rate annealing.
- Currently looking into SAC and its differences: which might be the order of operations, value loss weight, inconsistent learning rates of networks, frequency of gradient steps, and entropy coefficient optimization. Applied this on HalfCheetah-v4 as an initial test.

Plans for next weeks

- Run the experiments with wrappers for CleanRL again, due to inconsistent normalization of the states caused by reinitiating the normalization wrapper within evaluation. (Along with plotting the unnormalized rewards for Stable Baselines 3)
- Continue with experimenting on SAC: Ant-v4, possibly wrapped environments, reproduction of the SAC paper. Account difference in performance to difference in implementation.